



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2020 - 2021)

JEE (Advanced)

AIOT

19-09-2021

JEE(Main+Advanced) : Leader & Enthusiast Course

ANSWER KEY

PAPER-1

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4				
	A.	C	B	B	A				
SECTION-I (ii)	Q.	5	6	7	8	9	10	11	12
	A.	A,C	A,D	B,D	A,C,D	A,B,C,D	A,C	A,B	B,C,D
SECTION-II	Q.	1	2	3	4	5	6		
	A.	2.25	7.00	2.50	1.30	0.94	8.00		

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4				
	A.	D	A	A	C				
SECTION-I (ii)	Q.	5	6	7	8	9	10	11	12
	A.	A,B,C	A,B,C,D	A,B,C,D	C,D	A,B,D	B,C	A,B	A,D
SECTION-II	Q.	1	2	3	4	5	6		
	A.	9.00	5.00	6.00	7.00	7.00	9.00		

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4				
	A.	B	C	C	A				
SECTION-I (ii)	Q.	5	6	7	8	9	10	11	12
	A.	A,B	B,C	A,B,D	A,B,C	B,C,D	A,C,D	A,B,C	C,D
SECTION-II	Q.	1	2	3	4	5	6		
	A.	600.00	7.25	8.00	2.31	1.96	30.00		

HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

1. **Ans (C)**

When $v = 200$ m/s, $F_{\text{net}} = 0$

$$\therefore qv\beta = qE \Rightarrow B = E/200$$

And when $v = 400$ m/s, $F_{\text{net}} = 4 \times 10^{-19}$ N

$$\therefore qv\beta - qE = 4 \times 10^{-19}$$

$$e \left(400 \times \frac{E}{200} - E \right) = 4 \times 10^{-19}$$

$$1.6 \times 10^{-19} (E) = 4 \times 10^{-19}$$

$$E = \frac{4}{1.6} = 2.5 \text{ N/C}$$

2. **Ans (B)**

Resistance across capacitor

$$R = \frac{6 \times 12}{6 + 12} = 4\Omega \quad \tau (\text{constant}) = CR = 32 \mu\text{s}$$

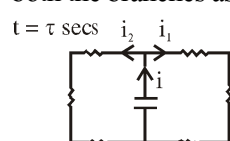
$$i = i_0 e^{-t/\tau} = \frac{v_c e^{-t/\tau}}{R} \quad \text{---(1)}$$

$$\text{where } v_c = \frac{160 \mu\text{s}}{8 \mu\text{F}} = 20 \text{ volts}$$

$$\text{In eq(1) } i = \frac{20 e^{-t/\tau}}{4} = 5 e^{-t/\tau}$$

At $t = \tau$ $i = 5 e^{-1} = 5/e$ Amp

This current is distributed in inverse of resistance in both the branches as shown



$$i_1 = \frac{5}{e} \left(\frac{6}{18} \right) i_1 = \frac{5}{3e}$$

(Current via 4Ω resistor)

3. **Ans (B)**

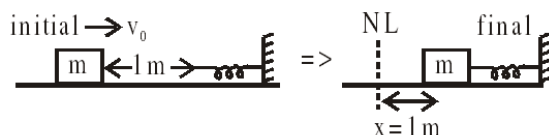
Let spring is compressed by x for the first time and block stops there for a moment. It will not return back if

$$kx \leq umg$$

for maximum value of x is

$$x = \frac{umg}{k} = \frac{1}{2} \frac{(2 \times 10)}{10} = 1\text{m}$$

Now work energy theorem from initial to final



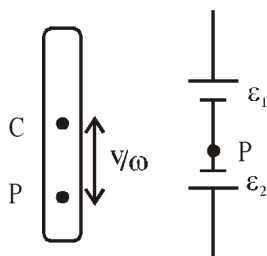
$$W_s + W_f = \Delta KE$$

$$-\frac{1}{2}Kx^2 - umg(x+1) = 0 - \frac{1}{2}mv_0^2, \quad [x = 1\text{m}]$$

After solving $v_0 = 5\text{ m/s}$

4. **Ans (A)**

Point p is at instantaneous rest



$$\epsilon_1 = |v_p - v_A| = \frac{1}{2}B\omega \left(\frac{l}{2} + \frac{v}{\omega} \right)^2$$

$$\epsilon_2 = |v_p - v_B| = \frac{1}{2}B\omega \left(\frac{l}{2} - \frac{v}{\omega} \right)^2$$

$$|v_A - v_B| = |\epsilon_1 - \epsilon_2|$$

On solving

$$|v_A - v_B| = B\ell v$$

SECTION-I (ii)

5. **Ans (A,C)**

There will be destructive interference, $2n_f d = m\lambda$

For complete destructive interference wave corresponding to ray (2) and (5) should have same amplitude.

$$-\frac{n_f - n_a}{n_f + n_a}a = -\frac{2n_a}{n_f + n_a} \frac{n_g - n_f}{n_g + n_f} \frac{2n_f}{n_f + n_a}a$$

$$\text{When } n_a \approx n_b, \frac{4n_a n_f}{(n_f + n_a)^2} \approx 1$$

$$\text{So, } n_f \approx \sqrt{n_a n_g}$$

6. **Ans (A,D)**

For (A), use the fact that potential at center of sphere, due to Q_2 will be exactly cancelled by charge $-Q_2$ induced on the surface of the cavity. Hence (A).

Potential at any point at surface of cavity is equal to potential at the centre of sphere. This will solve (B).

Potential outside the sphere cannot be found easily as the charge on the outer surface will be non-uniform. Hence C is incorrect as the given answer for uniform $Q_2 + Q_3$.

(D) can be calculated by making the net potential at the center of sphere zero and finding new charge on sphere.

7. **Ans (B,D)**

For A,B: Conserve angular momentum of each rod about points A and A' respectively. Similarly for C,D.

8. **Ans (A,C,D)**

The current is $i = \frac{2E}{R + r_1 + r_2}$. For power output to be zero, $i = \frac{E}{r_1}$. For power output to be maximum, $i = \frac{E}{2r_1}$

9. **Ans (A,B,C,D)**

Least count is $0.5/50 = 0.01\text{mm}$

Zero error is $5 \times 0.01 = 0.05\text{ mm}$.

Zero correction is negative of zero error.

The answer should be reported to two places after decimal in mm.

10. **Ans (A,C)**

$$r = \frac{r_0 n^2}{Z}$$

$$Z = 1$$

$$0.01 \times 10^{-3} = 0.53 \times 10^{-10} \times n^2$$

$$n^2 = \frac{1}{53} \times 10^7 = \frac{1000}{53} \times 10^4$$

$$n = 435$$

$$E = -\frac{E_0}{n^2} = -\frac{13.6\text{eV}}{(435)^2}$$

11. **Ans (A,B)**

$$\text{For B } \frac{2\pi}{w'} = \frac{2\pi}{w_s - w_e} = \frac{2\pi}{\frac{2\pi}{T_s} - \frac{2\pi}{T_e}} = \frac{T_s \times T_e}{(T_e - T_s)}$$

$$\text{For C } \frac{2\pi}{w'} = \frac{2\pi}{w_s + w_e} = \frac{T_s \times T_e}{(T_e + T_s)}$$

12. **Ans (B,C,D)**

$$d_{\text{block}} = \frac{1}{3 \times 10^{-4}} \text{ kg/m}^3 = \frac{10}{3} \times 10^3 \text{ kg/m}^3$$

$$B = 3 \times 10^{-4} \times 10^3 \times g = 3N$$

$$T_B = 10 - 3 = 7N$$

$$\text{Reading of B} = 14 + 3 = 17N;$$

$$\text{Reading of C} = [14 + (10 - 3) + 3] = 24N$$

SECTION-II

1. **Ans (2.25)**

The constant + oscillating current shows that

the L and C are in series, and this combination

is in parallel with $R = 250 \text{ ohm}$.

$$\frac{1}{Z} = \frac{1}{Z_1} + \frac{1}{Z_2}$$

$$\frac{1.25}{250} = \left| \frac{1}{250} + \frac{1}{i\omega L + \frac{1}{i\omega C}} \right|$$

$$= \left| \frac{1}{250} + \frac{i\omega C}{1 + \omega^2 LC} \right|$$

$$= \left| \frac{1}{250} + \frac{i2250C}{1 - 0.25} \right|$$

Hence:

$$\frac{1}{200^2} = \frac{1}{250^2} + \frac{16}{9} \times 2250^2 C^2$$

Hence $C = 10^{-6} \text{ F}$. Also:

$$4500^2 LC = 1$$

$$\text{Hence } L = \frac{4}{81} \text{ H.}$$

2. **Ans (7.00)**

Using the ideal gas equation, the given relation

can be expressed as $V = (nRT - \alpha)/P_0$. The

specific heat of the gas is

$$C = C_v + \frac{RT}{V} \frac{dV}{dT}$$

Here this simplifies to:

$$C = C_p + \frac{R\alpha}{nRT - \alpha}$$

The heat absorbed is:

$$Q = \int nC dT$$

$$= nC_p \Delta T + \alpha \ln(nRT - \alpha) \left[\frac{4\alpha}{nR} \right]$$

$$\text{Hence } \frac{Q}{\alpha} = \frac{2\gamma}{\gamma - 1} + \ln 3 = 5.90 + 1.10 = 7.00$$

3. **Ans (2.50)**

The intensity produced by the first slit alone is

$$10 \times \frac{2 \times 0.01}{2 \times 40} = 0.0025$$

By the second slit, it is 0.01 W/m^2 . The net path

$$\text{difference at P is } \frac{y d}{D - (\mu - 1) t}$$

$$= 3.33\lambda - 0.5t. \text{ Hence for minima we get } 3.33\lambda - 0.5t$$

$$= 2.5\lambda \Rightarrow t = 1.67\lambda = 1\mu\text{m. The intensity is}$$

$$0.0025 + 0.01 - 2\sqrt{0.0025 \times 0.01} = 0.0025 \text{ W/m}^2.$$

$$\text{Hence the product is } 25 \times 10^{-4} \times 10^{-6} = 2.50 \text{ nW/m.}$$

4. Ans (1.30)

$$\begin{aligned} PE &= \int dmgy \\ &= \int yg\mu\sqrt{(dx)^2 + (dy)^2} \\ &= \mu g \int y \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx \\ &= \mu gb \int \cosh \frac{x}{b} \sqrt{1 + \sinh^2 \frac{x}{b}} dx \\ &= \mu gb \int \frac{\cosh^2 x}{b} dx \\ &= \frac{\mu gb}{2} \int_{-\alpha}^{\alpha} 1 + \cosh \frac{2x}{b} dx \\ &= \frac{\mu gb}{2} \left(2a + 2 \times \frac{b}{2} \sinh \frac{2a}{b} \right) \\ &= \mu gb^2 \left(\frac{a}{b} + \frac{1}{2} \sinh \frac{2a}{b} \right) \\ &= 0.02 \times 10 \times 2 \times 2 \left(\frac{11}{16} + \frac{4 - 0.25}{4} \right) \\ &= 1.30 \text{ J} \end{aligned}$$

$$\Rightarrow x = 1.5 \text{ m}$$

5. Ans (0.94)

$$h = 160 \text{ m}$$

$$t_{10} = \sqrt{\frac{2h}{g}} = \sqrt{\frac{2 \times 160}{10}} = 4\sqrt{2} \text{ sec}$$

Velocity of system after collision in vertical direction $v_{10} = 30 \text{ m/s}$.

Velocity of system after collision in horizontal direction $v_2 = 1/4 \times 40 = 10 \text{ m/s}$

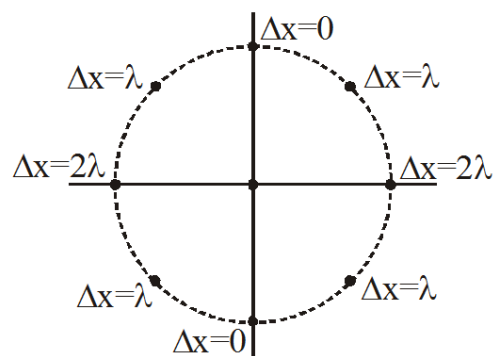
$$v_{10}^2 = 900 + 2 \times 10 \times 80$$

$$\Rightarrow v'_{10} = 50 \text{ m/s}.$$

$$t_{20} = \frac{50 - 30}{10} = 2 \text{ sec}$$

$$t_{10} = 4 \text{ sec}.$$

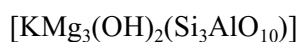
6. Ans (8.00)



PART-2 : CHEMISTRY

SECTION-II

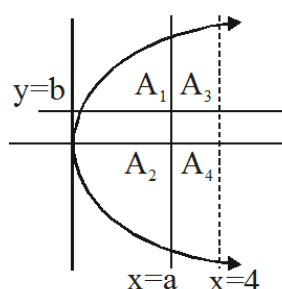
4. Ans (7.00)



PART-3 : MATHEMATICS

SECTION-I (i)

1. Ans (B)



$$A_3 = A_4 = \frac{8}{4}$$

$$\int_a^4 (\sqrt{x} - b) dx = \int_a^4 (b + \sqrt{x}) dx = \frac{8}{3}$$

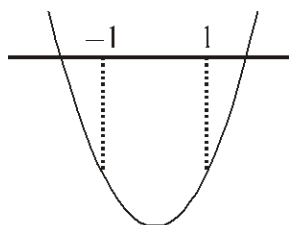
$$\therefore a^3 = 16, b = 0$$

2. Ans (C)

$$x^2 - 1 = 0 \rightarrow x = \pm 1$$

$$y = f(x) = x^2 + (3a - a^2)x - 3a^3$$

$$f(-1) < 0 \Rightarrow 1 - 3a + a^2 - 3a^3 < 0$$



$$\Rightarrow 3a^3 - a^2 + 3a - 1 > 0$$

$$\Rightarrow 3a(a^2 + 1) - 1(a^2 + 1) > 0$$

$$\Rightarrow (3a - 1)(a^2 + 1) > 0$$

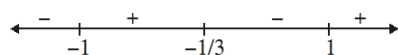
$$\therefore a > \frac{1}{3} \Rightarrow a \in \left(\frac{1}{3}, \infty\right)$$

$$f(1) < 0 \Rightarrow 1 + 3a - a^2 - 3a^3 < 0$$

$$\Rightarrow 3a^3 + a^2 - 3a - 1 > 0$$

$$\Rightarrow 3a(a^2 - 1) + (a^2 - 1) > 0$$

$$\Rightarrow (a - 1)(a + 1)(3a + 1) > 0$$



$$\therefore a \in \left(-1, -\frac{1}{3}\right) \cup (1, \infty)$$

Hence, common solution is $a \in (1, \infty)$

3. Ans (C)

$$T_{r+1} = {}^{2n}C_r (x \sin \alpha)^{2n-r} \left(\frac{\cos \alpha}{x}\right)^r$$

For term independent of x , $r = n$

$$\Rightarrow T_{n+1} = {}^{2n}C_n \left(\frac{\sin 2\alpha}{2}\right)^n \leq \frac{{}^{2n}C_n}{2^n}$$

4. Ans (A)

Let the line $y = mx + c$ containing the point

$$\left(\frac{t^3}{t-1}, \frac{t^2-3}{t-1}\right) \text{ where } t = a, b, c (a \neq b \neq c)$$

then

$$\frac{t^2-3}{t-1} = m \frac{t^3}{t-1} + c$$

$$\Rightarrow mt^3 - t^2 + ct + (3-c) = 0$$

This is a cubic equation in t having roots a, b

and c

$$\text{Hence, } a + b + c = \frac{1}{m}$$

$$ab + bc + ca = \frac{c}{m} \text{ and } abc = -\frac{3-c}{m}$$

$$\Rightarrow abc - \frac{c}{m} + \frac{3}{m} = 0$$

$$\Rightarrow abc - (ab + bc + ca) + 3(a + b + c) = 0$$

SECTION-I (ii)

5.

Ans (A,B)

$$(A) \lim_{x \rightarrow 0^-} \left[x + \frac{1}{2}\right] = 0, \lim_{x \rightarrow 0^+} \left[2x + \frac{1}{3}\right] = 0$$

$$\text{Hence, } \lim_{x \rightarrow 0} [f(x)] = 0$$

$$(B) \lim_{x \rightarrow 0^-} \left(x + \frac{1}{2}\right) = \frac{1}{2}, \lim_{x \rightarrow 0^+} \left(2x + \frac{1}{3}\right) = \frac{1}{3}$$

Since, $f(0^-) \neq f(0^+)$.

Therefore, $\lim_{x \rightarrow 0} f(x)$ does not exist.

(C) Since, $\lim_{x \rightarrow 0} f(x)$ does not exist so

$[\lim_{x \rightarrow 0} f(x)]$ does not exist.

$$(D) \lim_{x \rightarrow 0^-} \frac{[x + \frac{1}{2}]}{x} = 0, \lim_{x \rightarrow 0^+} \frac{[2x + \frac{1}{3}]}{x} = 0$$

$$\text{Therefore, } \lim_{x \rightarrow 0} \frac{[f(x)]}{x} = 0$$

6. **Ans (B,C)**

$$f(z) = z^4 + a_1 z^3 + a_2 z^2 + a_3 z + a_4$$

$$\therefore f(ki) = k^4 - a_1 k^3 i - a_2 k^2 + a_3 ki + a_4 = 0$$

$$\therefore k^4 - k^2 a_2 + a_4 = 0,$$

$$k^3 a_1 - k a_3 = 0 \Rightarrow \frac{a_3}{a_1 a_2} + \frac{a_1 a_4}{a_2 a_3} = 1$$

7. **Ans (A,B,D)**

The foot of perpendicular from the focus on any tangent to the parabola must lie on the tangent at the vertex to the parabola. Here, the foot of perpendicular from focus (4,2) on the given tangent is (1, -2) i.e. the point of tangency only hence, the given tangent is the tangent at the vertex (1,-2) of the parabola.

Hence, the foot of directrix P (h,k) will be

$$\begin{array}{ccc} & | & \\ \hline P(h,k) & A(1,-2) & S(4,2) \end{array}$$

$$\left(\frac{h+4}{2}, \frac{k+2}{2} \right) = (1, -2) \therefore (h, k) = (-2, -6)$$

$\therefore$ the equation of the directrix will be

$$3(x+2) + 4(y+6) = 0 \Rightarrow 3x + 4y + 30 = 0$$

On which the point (2, -9) lies.

$$\text{Also, the length of } LR = 4AS = 4\sqrt{(1-4)^2 + (-2-2)^2} = 20$$

8. **Ans (A,B,C)**

Intersection of line when both the planes are same

$$\frac{3}{3\beta^2 + 6(1-2\alpha) + 3} = \frac{-6}{6\alpha^2 + 6(1-2\beta) + 6}$$

$$\Rightarrow 2(\beta-1)^2 + 3(\alpha-2)^2 = 0 \Rightarrow \alpha = 2, \beta = 1$$

$\therefore (\alpha, \beta, 1)$ lies on the planes

$$2x + y + z = 4 \text{ (A)}$$

$$x + y - 1 = 2 \text{ (B) and } x - 2y = 0 \text{ (C)}$$

9. **Ans (B,C,D)**

$$\Rightarrow \lim_{n \rightarrow \infty} \frac{\int_0^{n\pi} \sin^4 x dx}{n\pi} = \lim_{n \rightarrow \infty} \frac{n \int_0^{\pi} \sin^4 x dx}{n\pi} = \frac{2}{\pi} \int_0^{\pi/2} \sin^4 x dx$$

$$= \frac{2}{\pi} \cdot \frac{3}{4} \cdot \frac{1}{2} \cdot \frac{\pi}{2} = \frac{3}{8}$$

10. **Ans (A,C,D)**

$$\frac{S}{S_1} = \frac{(BC)^2}{(KM)^2} \Rightarrow \sqrt{S} = \frac{BC}{KM} \cdot \sqrt{S_1} \dots (1)$$

$$\frac{S}{S_2} = \frac{(BC)^2}{(ML)^2} \Rightarrow \sqrt{S} = \frac{BC}{KM} \cdot \sqrt{S_1} \dots (2)$$

$$\frac{S}{S_2} = \frac{(BC)^2}{(MP)^2} \Rightarrow \sqrt{S} = \frac{BC}{NP} \cdot \sqrt{S_3} \dots (3)$$

adding eq(i), eq(ii) and eq(iii)

$$\sqrt{S} (KM + ML + NP) = BC (\sqrt{S_1} + \sqrt{S_2} + \sqrt{S_3})$$

$$\sqrt{S} (BN + PC + NP) = BC (\sqrt{S_1} + \sqrt{S_2} + \sqrt{S_3})$$

$$\sqrt{S} = \sqrt{S_1} + \sqrt{S_2} + \sqrt{S_3}$$

11. **Ans (A,B,C)**

$$xf(y) < yf(x) \forall 0 < x < y < 1$$

$$\Rightarrow \frac{f(y)}{y} < \frac{f(x)}{x} \forall x < y$$

$$\Rightarrow g(x) = \frac{f(x)}{x} \text{ is decreasing}$$

$$\Rightarrow g'(x) < 0 \Rightarrow xf'(x) < f(x)$$

$$\Rightarrow f'(x) < \frac{f(x)}{x} \forall x \in (0, 1)$$

$$\text{Hence, } f'(x) < g(x) \forall x \in (0, 1)$$

Hence, $f'(x)$ is less than minimum value of $g(x)$ which is $g(1)$

$$\therefore f'(x) < f(1)$$

$$xf'(x) < f(x) \Rightarrow \int_0^1 xf'(x) dx < \int_0^1 f(x) dx \dots (1)$$

$$\Rightarrow xf(x)|_0^1 - \int_0^1 f(x) dx < \int_0^1 f(x) dx$$

$$\Rightarrow f(1) < 2 \int_0^1 f(x) dx \dots (2)$$

12. Ans (C,D)

As M is a 2×2 symmetric matrix,

$$M = \begin{bmatrix} a & b \\ b & c \end{bmatrix}$$

where, $a, b, c, \in I$

The first column of M is $\begin{bmatrix} a \\ b \end{bmatrix}$ and second row of M is $[b \ c]$.

$$\text{If } \begin{bmatrix} a \\ b \end{bmatrix} = [b \ c]' = \begin{bmatrix} b \\ c \end{bmatrix}$$

$$\Rightarrow a = b = c$$

and hence M is not invertible

The second row of M is $[b \ c]$ and first column

is $\begin{bmatrix} a \\ b \end{bmatrix}$

If $[b \ c] = \begin{bmatrix} a \\ b \end{bmatrix}$ $[a \ b]$ we get $a = b = c$ and hence

M is not invertible

If M is a diagonal matrix with non zero entries, then

$$M = \begin{bmatrix} a & 0 \\ 0 & c \end{bmatrix} \text{ and } \det(M) = ac \neq 0$$

$\therefore M$ is invertible

If the product of main diagonal elements is not a perfect square, then $\det(M) = ac - b^2 \neq 0$

and hence M is invertible.

SECTION-II

1. **Ans (600.00)**

$$\Rightarrow \sum_{r=0}^n \frac{1}{(\sqrt{4r+1} + \sqrt{4r+5})(\sqrt[4]{4r+5} + \sqrt[4]{4r+1})} \cdot \frac{\sqrt[4]{4r+5} - \sqrt[4]{4r+1}}{\sqrt[4]{4r+5} - \sqrt[4]{4r+1}} = \frac{3}{2}$$

$$\Rightarrow \sum_{r=0}^n \frac{\sqrt[4]{4r+5} - \sqrt[4]{4r+1}}{4r+5 - 4r-1} = \frac{3}{2}$$

$$\Rightarrow \frac{1}{4}[(\sqrt[4]{5}-1)+(\sqrt[4]{9}-\sqrt[4]{5})+....+(\sqrt[4]{4n+5}-\sqrt[4]{4n+1})]=\frac{3}{2}$$

$$\Rightarrow \frac{1}{4}(\sqrt[4]{4n+5}-1) = \frac{3}{2} \Rightarrow 4n+5 = \left(\frac{3}{2} \times 4 + 1\right)^4 = 2401$$

$$\Rightarrow n = 599$$

$$\therefore \text{Number of terms} = n + 1 = 600$$

2. Ans (7.25)

$$2y \frac{dy}{dx} \sin x + y^2 \cos x = \sin x \cos x$$

$$\Rightarrow \frac{d}{dx} (\sin x \cdot y^2) = \sin x \cos x$$

$$\Rightarrow \sin x \cdot y^2 = \int \sin x \cos x \, dx$$

$$\Rightarrow \sin x \cdot y^2 = \frac{\sin^2 x}{2} + \frac{c}{2}$$

$$\Rightarrow 2 \sin x \cdot y^2 = \sin^2 x + c$$

Substituting $x = \frac{\pi}{2}, y = 2$

$$\Rightarrow 2 \cdot \sin \frac{\pi}{2} \cdot 2^2 = \sin^2 \frac{\pi}{2} + c$$

$$\therefore c = 8 - 1 = 7$$

$$\therefore 2 \sin x \cdot y^2 = \sin^2 x + 7$$

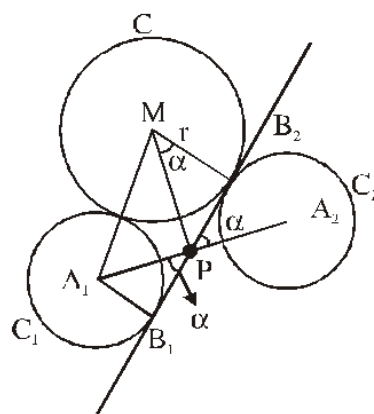
$$\therefore y^2 = \frac{1}{2}\sin x + \frac{7}{2}\operatorname{cosec} x = f(x)$$

$$\therefore f\left(\frac{\pi}{6}\right) = \frac{1}{2} \cdot \frac{1}{2} + \frac{7}{2} \cdot 2 = 7.25$$

3. Ans (8.00)

Let A_1 , A_2 and M be the centres of the circles C_1 , C_2 and C respectively. Let the common tangent through P to C_1 and C touch C_1 at B_1 , C at B , and C_2 also at B_2 .

From right angled triangle $A_1 B_1 P$



If $\angle A_1 P B_1 = \alpha$, $\sin \alpha = \frac{A_1 B_1}{A_1 P} = \frac{1}{3}$

$$\Rightarrow \cos \alpha = \frac{2\sqrt{2}}{3} \Rightarrow PB_1 = 2\sqrt{2} = PB_2$$

From triangle MPB,

$$\tan \alpha = \frac{PB_2}{MB_2} = \frac{2\sqrt{2}}{r}$$

$$\Rightarrow \frac{1}{2\sqrt{2}} = \frac{2\sqrt{2}}{r} \Rightarrow r = 8$$

4. Ans (2.31)

$$\begin{aligned}
 \text{Required Probability} &= \frac{{}^6C_3 \cdot {}^6C_3}{{}^{12}C_6} \\
 P &= \frac{6 \cdot 5 \cdot 4}{3 \cdot 2 \cdot 1} \cdot \frac{6 \cdot 5 \cdot 4}{3 \cdot 2 \cdot 1} \cdot \frac{6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1}{12 \cdot 11 \cdot 10 \cdot 9 \cdot 8 \cdot 7} \\
 P &= \frac{100}{231} \therefore \frac{1}{p} = \frac{231}{100} = 2.31
 \end{aligned}$$

5. Ans (1.96)

$$\begin{aligned}
 r_1 &= 2r_2 = 3r_3 \\
 \Rightarrow \frac{r_1}{6} = \frac{r_2}{3} = \frac{r_3}{2} &\Rightarrow \frac{s \tan(\frac{A}{2})}{6} = \frac{s \tan(\frac{B}{2})}{3} = \frac{s \tan(\frac{C}{2})}{2} \\
 \Rightarrow \frac{\tan \frac{A}{2}}{6} &= \frac{\tan \frac{B}{2}}{3} = \frac{\tan \frac{C}{2}}{2} \\
 &= \frac{\sqrt{\tan \frac{A}{2} \cdot \tan \frac{B}{2} + \tan \frac{B}{2} \cdot \tan \frac{C}{2} + \tan \frac{C}{2} \cdot \tan \frac{A}{2}}}{\sqrt{6 \times 3 + 3 \times 2 + 2 \times 6}} \\
 &= \frac{1}{6} \\
 \therefore \tan \frac{A}{2} &= 1, \tan \frac{B}{2} = \frac{1}{2} \text{ \& } \tan \frac{C}{2} = \frac{1}{3} \\
 \therefore \angle A &= \frac{\pi}{2}, \sin B = \frac{2 \cdot \frac{1}{2}}{1 + (\frac{1}{2})^2} = \frac{4}{5} \quad \& \\
 \sin C &= \frac{2 \cdot \frac{1}{3}}{1 + (\frac{1}{3})^2} = \frac{3}{5} \\
 \therefore \cos B &= \frac{3}{5} \text{ and } \cos C = \frac{4}{5} \\
 \therefore (\sin B + \sin C)(\cos B + \cos C) &= \left(\frac{3+4}{5}\right) \left(\frac{4+3}{5}\right) = \frac{196}{100} = 1.96
 \end{aligned}$$

6. Ans (30.00)

$$\begin{aligned}
 \begin{bmatrix} \vec{a} & \vec{b} & \vec{c} \end{bmatrix}^2 &= \begin{vmatrix} \vec{a} \cdot \vec{a} & \vec{a} \cdot \vec{b} & \vec{a} \cdot \vec{c} \\ \vec{b} \cdot \vec{a} & \vec{b} \cdot \vec{b} & \vec{b} \cdot \vec{c} \\ \vec{c} \cdot \vec{a} & \vec{c} \cdot \vec{b} & \vec{c} \cdot \vec{c} \end{vmatrix} = \begin{vmatrix} 9 & 6 & -1 \\ 6 & 16 & 6 \\ -1 & 6 & 36 \end{vmatrix} \\
 &= 9(576 - 36) - 6(216 + 6) \\
 &\quad - (36 + 16) \\
 &= 4860 - 1332 - 52 = 3476 \\
 \therefore \text{Volume of tetrahedron} &= V = \frac{1}{6} [\vec{a} \vec{b} \vec{c}] \\
 \therefore V^2 &= \frac{1}{36} \cdot [\vec{a} \vec{b} \vec{c}]^2 = \frac{1}{36} \times 3476 \\
 &= \frac{869}{9} = \frac{11 \times 79}{3^2} = \frac{pq}{r^2} \\
 \therefore \frac{(p+q)}{r} &= \frac{11+79}{3} = 30
 \end{aligned}$$

JEE(Main + Advanced) : Leader & Enthusiast Course
ANSWER KEY
PAPER-2
PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4	5	6	7	8
	A.	A,C	A,B,D	D	A,B	A,B,C	A,B,D	A,B	A,B,D
SECTION-I (ii)	Q.	9	10	11	12				
	A.	B	A	C	B				
SECTION-II	Q.	1	2	3	4	5	6		
	A.	121.69 to 121.70	120.00	4.80	0.50	8.00	1.33		

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4	5	6	7	8
	A.	B,C,D	B	A,B,C	B	B	A,D	B,C,D	B,C,D
SECTION-I (ii)	Q.	9	10	11	12				
	A.	A	C	A	A				
SECTION-II	Q.	1	2	3	4	5	6		
	A.	1.00	3.00	7.00	8.00	4.00	14.55		

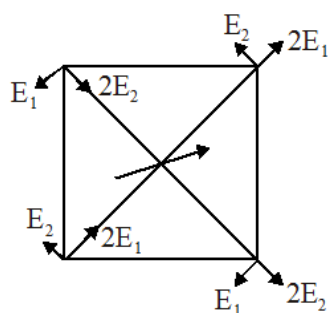
PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4	5	6	7	8
	A.	A,B,C	A,B,D	A,B,C,D	B,D	A,C	B,C	B,C	C
SECTION-I (ii)	Q.	9	10	11	12				
	A.	A	C	B	D				
SECTION-II	Q.	1	2	3	4	5	6		
	A.	3.00	0.00	7.00	14.00	0.50	0.00		

HINT – SHEET
PART-1 : PHYSICS
SECTION-I (i)

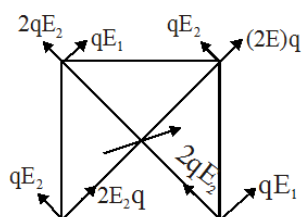
1. Ans (A,C)

Electric field diagram due to dipole



$$E_1 = \frac{kP \cos \theta}{r^3}; E_2 = \frac{kP \sin \theta}{r^3}$$

Force diagram on the system



$$\begin{aligned} \text{Net force} &= \sqrt{(6qE_1)^2 + (6qE_2)^2} \\ &= \frac{6kPq}{r^3} \sqrt{\sin^2 \theta + \cos^2 \theta} \quad \text{Net force} = \frac{6kPq}{r^3} \end{aligned}$$

The net torque on the system = 0

2. Ans (A,B,D)

The height on both sides will be obviously different as the materials are different and hence the wetting angles will also be different.

According to the law of conservation of energy, the cross will not rotate. The components of the force of surface tension are balanced by the forces of hydrostatic pressure.

3. Ans (D)

$$m_{\alpha} = 4 \times 1 \text{ a.m.u.} = 4 \times 1.66 \times 10^{-27} \text{ kg}$$

$$\frac{1}{2} m_{\alpha} v^2 = 4.78 \text{ MeV} = 4.78 \times 1.6 \times 10^{-19} \times 10^6 \text{ J}$$

$$\Rightarrow v = 1.55 \times 10^7 \text{ m/s}$$

Total energy evolved in the reaction

$$\Rightarrow \varepsilon = KE_{\alpha} + KE_{\text{residual nucleus}}$$

Using momentum conservation law,

$m_{\alpha} V_{\alpha} = MV$ where M and V are mass and velocity of residual nucleus.

$$\therefore m_{\alpha}^2 V_{\alpha}^2 = M^2 V^2$$

$$\therefore \varepsilon = \frac{1}{2} m_{\alpha} v_{\alpha}^2 + \frac{m_{\alpha}}{M} \frac{1}{2} m_{\alpha} v_{\alpha}^2$$

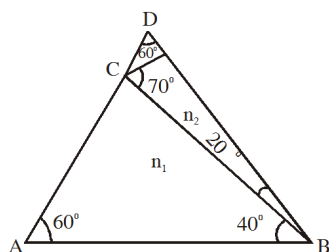
$$= \left(\frac{1}{2} m_{\alpha} v_{\alpha}^2 \right) \left[\frac{M + m_{\alpha}}{M} \right]$$

$$= (4.78 \text{ MeV}) \left[\frac{226}{222} \right] = 4.87 \text{ MeV}$$

4. Ans (A,B)

(I) For given condition

$$n_1 = 1.20 + \frac{10.8 \times 10^4}{\lambda^2} \quad \& \quad n_2 = 1.45 + \frac{1.80 \times 10^4}{\lambda^2}$$



For given condition $n_1 = n_2$

$$1.20 + \frac{10.8 \times 10^4}{\lambda^2} = 1.45 + \frac{1.80 \times 10^4}{\lambda^2}$$

$$\frac{9 \times 10^4}{\lambda_0^2} = 0.25 \Rightarrow \lambda_0^2 = 36 \times 10^4$$

$$n_2 = 1.20 + \frac{10.8 \times 10^4}{36 \times 10^4} = 1.5$$

$$n_1 = n_2 = 1.5$$

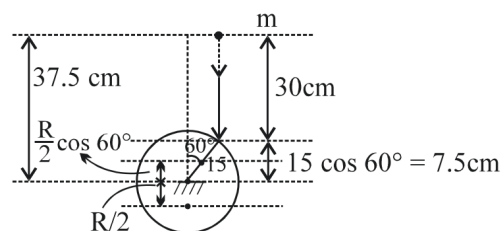
(II) for $\delta = \delta_m$, $r_1 = r_2 = \frac{A}{2}$

$$n = \frac{\sin i}{\sin r_1} \Rightarrow n = \frac{\sin i}{\sin \frac{A}{2}}$$

$$\sin i = 1.5 \sin 30^\circ$$

$$i = \sin^{-1}(0.75)$$

5. Ans (A,B,C)



$$v_m = \sqrt{2gh} = \sqrt{6} \text{ m/s}$$

applying conservation of angular momentum at the time of collision

$$L_i = L_f$$

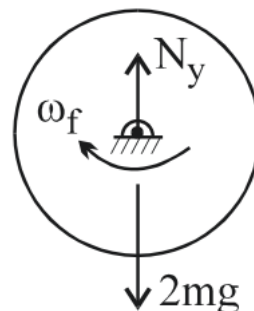
$$m(\sqrt{6}) \frac{15}{100} \sin 60^\circ = \left(\frac{mR^2}{2} + mR^2 \right) \omega_i$$

$$\omega_i = \frac{20\sqrt{2}}{3} \text{ rad/s}$$

Now, applying conservation of energy initial and final position (i.e. when centre of mass is at lowest position)

$$2mg \left(\frac{R}{2} + \frac{R}{2} \cos 60^\circ \right) = \frac{1}{2} I (\omega_f^2 - \omega_i^2)$$

$$\omega_f = \frac{20}{3} \sqrt{5} \text{ rad/s}$$



$$N_y - 2mg = 2m \left(\frac{R}{2} \right) \omega_f^2 \text{ on solving}$$

$$N_y = \frac{16mg}{3}$$

6. Ans (A,B,D)

Only the lengths x and y will experience a magnetic force. The magnetic force will act at the respective mid-points of the lengths x & y . Now find the magnetic force acting on both the sides and calculate torque about the centre of the square loop. The current is already specified in the diagram in the clockwise direction.

7. **Ans (A,B)**

For chamber:

$$\frac{dQ}{dt} = k(\theta_1 - \theta_0) = k(\theta_2 - \theta_0)$$

$$\Rightarrow \theta_1 = \theta_2$$

For heater:

$$\frac{dQ}{dt} = e_1 A \sigma (T_1^4 - \theta_1^4) = e_2 A \sigma (T_2^4 - \theta_2^4)$$

$$e_1 > e_2 \Rightarrow T_1 < T_2$$

8. **Ans (A,B,D)**

The magnetic field induction at point R due to current in wires P and Q is given by

$$B = B_1 + B_2 = \left(\frac{\mu_0 I_1}{2\pi d_1} + \frac{\mu_0 I_2}{2\pi d_2} \right)$$

$$= 2 \times 10^{-7} \left(\frac{2.5}{5} + \frac{1}{2} \right) \hat{j} \text{ along negative Y-axis}$$

where $\frac{\mu_0}{2\pi} = 2 \times 10^{-7}$ and $\hat{j}$ being the unit vector along Y-axis.

$$= -10^{-7} (1 + I) \hat{j} \quad (\because I_2 = I)$$

We know that $F = qvB$

$$= (-1.6 \times 10^{-19}) \times (4 \times 10^5 \hat{i}) \times \{-10^{-7} (1 + I) \hat{j}\}$$

$$= 6.4 \times 10^{-21} (1 + I) \hat{k} \quad (\because \hat{i} \times \hat{j} = \hat{k})$$

According to the problem

$$6.4 \times 10^{-21} (1 + I) = 3.2 \times 10^{-20}$$

$$\therefore I = 4 \text{ ampere}$$

Let the third wire be placed at a distance x on the left of R. Suppose the wire carries a current upwards perpendicular to the plane of paper. Now

$$B = B_1 + B_2 + B_3$$

$$= \frac{\mu_0}{2\pi} \left(\frac{I_1}{d_1} + \frac{I_2}{d_2} + \frac{I_3}{x} \right) = 0$$

($\because$ Directions of I_1 and I_2 are along negative Z-direction and direction of I_3 is along positive Z-direction)

$$\text{or } B = \frac{2.5}{5} + \frac{4}{2} - \frac{2.5}{x} = 0$$

$$\therefore x = 1 \text{ metre}$$

Note : If the direction of current in the third wire will be inward (negative Z-direction), then

$$B = \frac{\mu_0}{2\pi} \left(\frac{2.5}{5} + \frac{4}{2} + \frac{2.5}{x} \right) = 0$$

$$\therefore x = -1 \text{ metre}$$

Hence the third wire may be placed at ± 1 metre from R along X-axis.

SECTION-I (ii)

9. **Ans (B)**

At $t = 0$ inductor behaves as an insulator and

$$\text{hence } i = \frac{V}{R_{\text{net}}} = \frac{V}{7R/2}$$

now potential across inductor is

$$V_i = \left(\frac{2V}{7R} \right) 3R = \frac{6V}{7}$$

10. **Ans (A)**

When switch was opened after a long time the

current in inductor remains same at $t = 0$ also

after a long time inductor behaves like a

conductor.

$$\therefore i = \frac{V}{2R} \text{ (Current given by battery)}$$

This current divides equally in two branches

having resistance '3R' each

$$\therefore i = \frac{V}{4R} \text{ (is inductor branch)}$$

11. **Ans (C)**

In case (A) and (D) particles are at their

amplitude hence their K.E. = 0

$$\text{for B.} = \int_0^1 \frac{1}{2} \mu dx (\omega a \sin kx)^2$$

12. **Ans (B)**

For Option D

$$\text{At } x = \frac{L}{12} \text{ and } t = 0$$

$$y = a \sin \frac{2\pi}{L} \frac{L}{12} = \frac{a}{2}$$

SECTION-II

1. **Ans (121.69 to 121.70)**

Energy conservation gives

$$\text{Energy of H atom in } n^{\text{th}} \text{ state} + \frac{hc}{\lambda_1} = 12.2 \text{ eV}$$

$$\Rightarrow -\frac{13.6}{n^2} \text{ eV} + \frac{hc}{\lambda_1} = 12.2 \text{ eV}$$

$$\Rightarrow -\frac{13.6}{n^2} + \frac{6.62 \times 10^{-34} \times 3 \times 10^8}{796 \times 10^{-10} \times 1.6 \times 10^{-19}} = 12.2$$

$$\Rightarrow \frac{13.6}{n^2} = 15.6 - 12.2 = 3.4 \text{ eV} \therefore n = 2$$

Hydrogen atom formed is in 1st excited state.

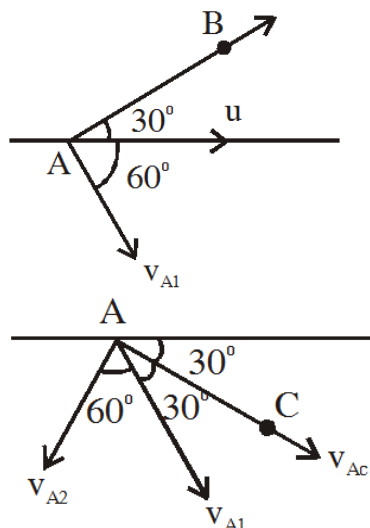
$$\therefore \frac{hc}{\lambda_2} = 10.2 \text{ eV}$$

$$\lambda_2 = \frac{6.62 \times 10^{-34} \times 3 \times 10^8}{10.2 \times 1.6 \times 10^{-19}}$$

2. **Ans (120.00)**

In an oblique elastic collision of two identical balls, the two will move perpendicular to each other after collision. The figure shows the centres of the balls. After first collision, B moves along the line of impact and A moves perpendicular to that. Then A hits C with velocity v_{A1} which makes 30° angle with line of impact. C goes along the impact and A is deflected perpendicular to that.

v_{A3} makes 120° with original direction of motion.



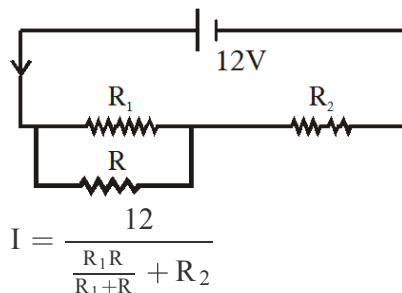
3. **Ans (4.80)**

The reading of voltmeter is absolutely correct when it is connected across the battery.

It means the emf of the cell is = 12 volts

Let the resistance of the voltmeter be R.

The current I is



$\therefore$ Voltmeter reading = voltage drop across

R_1 (or R)

$$\Rightarrow 4 = \frac{12}{\frac{R_1 R}{R_1 + R} + R_2} \cdot \frac{R_1 R}{R_1 + R}$$

$$\therefore 1 + \frac{R_2 (R_1 + R)}{R_1 R} = 3$$

$$\Rightarrow \frac{R_2 (R_1 + R)}{R_1 R} = 2$$

$$\Rightarrow \frac{R_2}{R} + \frac{R_2}{R_1} = 2 \quad \dots(1)$$

Similarly

$$V_2 = \frac{12}{\frac{R_2 R}{R_2 + R} + R_1} \cdot \frac{R_2 R}{R_2 + R}$$

$$\Rightarrow 6 = \frac{12}{1 + \frac{R_1 (R_2 + R)}{R_2 R}}$$

$$\Rightarrow \frac{R_1 R_2 + R_1 R}{R_2 R} = 1$$

$$\Rightarrow \frac{R_1}{R} + \frac{R_1}{R_2} = 1 \quad \dots(2)$$

Eliminating R between (1) and (2)

$$\frac{R_1}{R_2} = \frac{2}{3}$$

The actual voltage across R_1 is

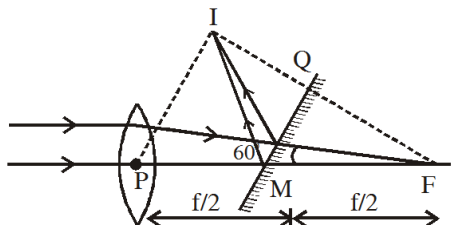
$$V_1 = \frac{12 \cdot R_1}{R_1 + R_2} = \frac{12}{1 + \frac{R_2}{R_1}} = \frac{12}{1 + \frac{3}{2}}$$

$$= \frac{24}{5} = 4.8 \text{ Volt}$$

4. Ans (0.50)

Focus point F acts as object for the plane mirror.
Its image is formed at I.

From geometry, $\angle IMG = 60^\circ$ and $MF = MI = \frac{f}{2}$



In $\triangle PIM$

$$IM = \frac{f}{2}$$

$$PM = \frac{f}{2}$$

$$IM = PM$$

and $\angle PMI = 60^\circ$

Hence, $\triangle PMI$ is equilateral.

$$PI = \frac{f}{2}$$

5. Ans (8.00)

$$E_i = -\frac{GMm}{2r} = -\frac{GMm}{8R}; E_f = -\frac{GMm}{2R}$$

Loss in energy

$$\Delta E = -\frac{GMm}{8R} - \left(-\frac{GMm}{2R} \right) = \frac{3GMm}{8R}$$

$$\therefore \eta t = \frac{3GMm}{8R}$$

$$t = \frac{3GMm}{8\eta R}$$

6. Ans (1.33)

$$I_A = \frac{7}{12}Ma^2$$

$$\therefore T = 2\pi \sqrt{\frac{\frac{7}{12}Ma^2}{Mg \frac{a}{\sqrt{2}}}} = 2\pi \sqrt{\frac{7}{6\sqrt{2}} \frac{a}{g}}$$

PART-3 : MATHEMATICS

SECTION-I (i)

1. Ans (A,B,C)

$$[\bar{a} \times \bar{p} \bar{b} \times \bar{q} \bar{c} \times \bar{r}] + [\bar{a} \times \bar{q} \bar{b} \times \bar{r} \bar{c} \times \bar{p}] + [\bar{a} \times \bar{r} \bar{b} \times \bar{p} \bar{c} \times \bar{q}]$$

$$[\bar{a} \times \bar{p} \bar{b} \times \bar{q} \bar{c} \times \bar{r}]$$

$$= (\bar{a} \times \bar{p}) \cdot \{ (\bar{b} \times \bar{q}) \times (\bar{c} \times \bar{r}) \}$$

$$= \{ \bar{a} \times \bar{p} \} \cdot \{ [\bar{b} \bar{q} \bar{r}] \bar{c} - [\bar{b} \bar{q} \bar{c}] \bar{r} \}$$

$$[\bar{a} \times \bar{p} \bar{b} \times \bar{q} \bar{c} \times \bar{r}]$$

$$= [\bar{a} \bar{p} \bar{c}] [\bar{b} \bar{q} \bar{r}] - [\bar{a} \bar{p} \bar{r}] [\bar{b} \bar{q} \bar{c}]$$

...(i)

$$[\bar{a} \times \bar{q} \bar{b} \times \bar{r} \bar{c} \times \bar{p}]$$

$$= \{ (\bar{a} \times \bar{q}) \times (\bar{b} \times \bar{r}) \} \cdot (\bar{c} \times \bar{p})$$

$$= [\bar{b} \bar{r} \bar{a}] [\bar{q} \bar{c} \bar{p}] - [\bar{b} \bar{r} \bar{q}] [\bar{a} \bar{c} \bar{p}]$$

...(ii)

$$[\bar{a} \times \bar{r} \bar{b} \times \bar{p} \bar{c} \times \bar{q}]$$

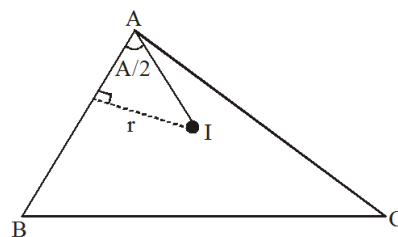
$$= (\bar{a} \times \bar{r}) \cdot \{ (\bar{b} \times \bar{p}) \times (\bar{c} \times \bar{q}) \}$$

$$= [\bar{c} \bar{q} \bar{b}] [\bar{a} \bar{r} \bar{p}] - [\bar{a} \bar{r} \bar{b}] [\bar{c} \bar{q} \bar{p}]$$

...(iii)

Now (i) + (ii) + (iii)

2. Ans (A,B,D)



$$AI = r \csc \frac{A}{2},$$

$$BI = r \csc \frac{B}{2},$$

$$CI = r \csc \frac{C}{2}$$

$$(AI)(BI)(CI) = \frac{r^3}{\sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2}}$$

$$= \frac{r^3}{r} \times 4R = 4r^2R$$

$$\Rightarrow a = 4, b = 2, c = 1$$

3. Ans (A,B,C,D)

$$P(A) = \frac{1}{2}, P(B) = \frac{1}{5}$$

$$P(A \cup B) = \frac{1}{2} + \frac{1}{5} - \frac{1}{2} \times \frac{1}{5} = \frac{3}{5}$$

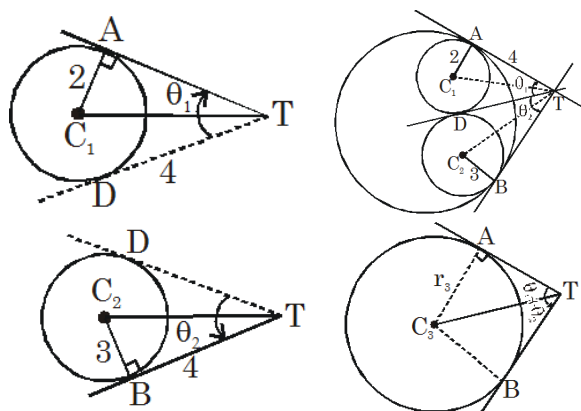
$$P\left(\frac{A}{B}\right) = P(A) = \frac{1}{2}$$

$$P\left(\frac{A}{A \cup B}\right) = \frac{P(A \cap (A \cup B))}{P(A \cup B)} = \frac{\frac{1}{2}}{\frac{3}{5}} = \frac{5}{6}$$

$$P(A \cap B/A' \cup B') = \frac{P((A \cap B) \cap (A' \cup B'))}{P(A' \cup B')} = 0$$

4. Ans (B,D)

AT and BT are radical axis of C_3 and C_1 ; C_3 and C_2 respectively



$\therefore T$ is radical centre $\Rightarrow$ radical axis of C_1 and C_2 i.e., common tangent passes through T .

$$TA = TB = TD = 4 \quad \tan \frac{\theta_1}{2} = \frac{2}{4} = \frac{1}{2},$$

$$\tan \frac{\theta_2}{2} = \frac{3}{4}, \quad (\angle ATD = \theta_1, \angle BTD = \theta_2)$$

$$r_3 = TA \tan \frac{\theta_1 + \theta_2}{2} = 4 \left(\frac{\frac{1}{2} + \frac{3}{4}}{1 - \frac{1}{2} \cdot \frac{3}{4}} \right) = 8$$

Circumcircle of $\triangle TAB$ will pass through C_3 has TC_3 as diameter

$$\Rightarrow \text{Area} = \pi \left(\frac{TC_3}{2} \right)^2, \quad TC_3 = \frac{TA}{\cos \left(\frac{\theta_1 + \theta_2}{2} \right)} = \frac{4}{1/\sqrt{5}} = 4\sqrt{5}$$

$$\text{Area} = \pi (2\sqrt{5})^2 = 20\pi$$

$$\left. \begin{array}{l} C_1C_3 = 8 - 2 = 6 \\ C_2C_3 = 8 - 3 = 5 \end{array} \right\} \Rightarrow C_1C_3 - C_2C_3 = 1$$

5. Ans (A,C)

$$\frac{dy}{dx} + Py = Q \Rightarrow \frac{dy_1}{dx} + Py_1 = Q \text{ and}$$

$$\frac{dy_2}{dx} + Py_2 = Q$$

$$\Rightarrow \frac{d}{dx}(y_1 - y_2) + P(y_1 - y_2) = 0$$

$$\Rightarrow \frac{d(y_1 - y_2)}{y_1 - y_2} = -P \, dx$$

$$\& \frac{d(y - y_1)}{dx} + P(y - y_1) = 0$$

$$\Rightarrow \frac{d(y - y_1)}{(y - y_1)} = -P \, dx$$

$$\Rightarrow \int \frac{d(y - y_1)}{y - y_1} = \int \frac{d(y_1 - y_2)}{y_1 - y_2}$$

$$\Rightarrow \ln(y - y_1) = \ln(y_1 - y_2) + \ln C$$

$$\Rightarrow y - y_1 = C(y_1 - y_2)$$

$$\Rightarrow y = y_1 + C(y_1 - y_2)$$

$$y = (1 + C)y_1 + (-C)y_2$$

$$\Rightarrow \alpha = 1 + C, \beta = -C$$

$$\alpha + \beta = 1$$

6. Ans (B,C)

$$\frac{n(n+1)}{2} - [n-1+n] \leq \text{sum of remaining}$$

$$'n-2' \text{ terms} \leq \frac{n(n+1)}{2} - (1+2)$$

$$\text{sum of remaining} = x_1 + x_2 + \dots + x_{n-2} = \frac{105}{4}(n-2)$$

where $n-2$ is multiple of 4.

$$\frac{n(n+1) - 2(2n-1)}{2} \leq \frac{105}{4}(n-2) \leq \frac{n^2 + n - 6}{2}$$

$$\Rightarrow \frac{(n-1)(n-2)}{2} \leq \frac{105}{4}(n-2) \& \frac{105}{4}(n-2) \leq \frac{(n+3)(n-2)}{2}$$

$$\Rightarrow 49.5 \leq n \leq 50.5$$

Hence, $n = 50$ is the only possibility

$$\text{Now, } \frac{\frac{50 \times 51}{2} - (k + k + 1)}{48} = \frac{105}{4}$$

$$\Rightarrow k = 7$$

7. Ans (B,C)

$$\begin{aligned} I - BA &= B [IB^{-1} - A] \\ &= B [IB^{-1} - ABB^{-1}] \\ &= B (I - AB) B^{-1} \\ |I - BA| &= |B (I - AB) B^{-1}| \\ &= |B| |I - AB| |B^{-1}| \\ &= |B| |I - AB| \frac{1}{|B|} = |I - AB| \neq 0 \\ \text{Again, } I - BA &= B (I - AB) B^{-1} \\ (I - BA)^{-1} &= \{B (I - AB) B^{-1}\}^{-1} \\ &= (B^{-1})^{-1} (I - AB)^{-1} B^{-1} \\ &= B(I - AB)^{-1} IB^{-1} \\ &= B(I - AB)^{-1} \left(\boxed{I - AB} + AB \right) B^{-1} \\ B(I - AB)^{-1} (I - AB) B^{-1} &+ B(I - AB)^{-1} ABB^{-1} \\ I + B(I - AB)^{-1} A & \end{aligned}$$

8. Ans (C)

$$f'(x) = 3(x^2 - 2ax + a^2 - 1)$$

Clearly roots of $f'(x) = 0$ must be distinct and lie in the interval $(-2, 4)$.

$$\therefore D > 0 \Rightarrow a \in \mathbb{R} \quad \dots\dots(1)$$

$$f'(-2) > 0 \Rightarrow a^2 + 4a + 3 > 0$$

$$\Rightarrow a < -3 \text{ or } a > -1 \quad \dots\dots(2)$$

$$f'(4) > 0 \Rightarrow a^2 - 8a + 15 > 0$$

$$\Rightarrow a > 5 \text{ or } a < 3 \quad \dots\dots(3)$$

$$\text{and } -2 < -\frac{B}{2A} < 4 \Rightarrow -2 < a < 4 \quad \dots\dots(4)$$

From (1), (2), (3) & (4), $a \in (-1, 3)$

SECTION-I (ii)

9. Ans (A)

$$(I) [\vec{a} \vec{b} \vec{c}]^2 = \begin{vmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 2 & -1 \end{vmatrix}^2 = 16$$

$$\begin{aligned} (II) \vec{n} &= \vec{r}_2 \times \vec{r}_3 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 3 & 4 \\ 2 & 1 & -2 \end{vmatrix} \\ &= \hat{i}(-6-4) - \hat{j}(-2-8) + \hat{k}(1-6) \\ &= -10\hat{i} + 10\hat{j} - 5\hat{k} \Rightarrow \hat{n} = \frac{2\hat{i} - 2\hat{j} + \hat{k}}{3} \end{aligned}$$

$$\begin{aligned} \therefore d &= |\text{Projection of } \overline{OP} \text{ on } \vec{n}| \\ &= \left| \frac{(3\hat{i} - 2\hat{j} - \hat{k}) \cdot (2\hat{i} - 2\hat{j} + \hat{k})}{3} \right| \\ &= \frac{6+4-1}{3} = 3 \text{ units} \end{aligned}$$

$$\begin{aligned} (III) L_1 : \frac{x-0}{1} &= \frac{y+1}{1} = \frac{z-0}{1} = \lambda; \\ L_2 : \frac{x+1}{2} &= \frac{y-0}{1} = \frac{z-0}{1} = \mu \end{aligned}$$

Hence any point on L_1 and L_2 can be

$$(\lambda, \lambda-1, \lambda) \text{ and } (2\mu-1, \mu, \mu)$$

$$\therefore \frac{2\mu-1-\lambda}{2} = \frac{\mu-\lambda+1}{1} = \frac{\mu-\lambda}{1}$$

solving $\mu = 1$ and $\lambda = 3$

$$A = (3, 2, 3) \text{ and } B = (1, 1, 1)$$

$$\Rightarrow AB = \sqrt{4+1+4} = 3$$

$$(IV) V_\ell = \frac{1}{6} [\vec{a} \vec{b} \vec{c}]; \quad V_s = \frac{1}{6} \cdot \frac{1}{27} [\vec{a} \vec{b} \vec{c}]$$

$$\text{Hence } \frac{V_s}{V_\ell} = \frac{1}{27} = \frac{m}{n} \text{ or } \frac{n}{27} = \frac{m}{1} = k$$

$\therefore m$ and n are relatively prime

$$\Rightarrow k = 1, (m+n) = 28$$

10. Ans (C)

$$(I) [\vec{a} \vec{b} \vec{c}]^2 = \begin{vmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 2 & -1 \end{vmatrix}^2 = 16$$

$$(II) \vec{n} = \vec{r}_2 \times \vec{r}_3 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 3 & 4 \\ 2 & 1 & -2 \end{vmatrix}$$

$$= \hat{i}(-6-4) - \hat{j}(-2-8) + \hat{k}(1-6)$$

$$= -10\hat{i} + 10\hat{j} - 5\hat{k} \Rightarrow \hat{n} = \frac{2\hat{i} - 2\hat{j} + \hat{k}}{3}$$

$$\therefore d = |\text{Projection of } \vec{OP} \text{ on } \vec{n}|$$

$$= \left| \frac{(3\hat{i} - 2\hat{j} - \hat{k}) \cdot (2\hat{i} - 2\hat{j} + \hat{k})}{3} \right|$$

$$= \frac{6+4-1}{3} = 3 \text{ units}$$

$$(III) L_1 : \frac{x-0}{1} = \frac{y+1}{1} = \frac{z-0}{1} = \lambda;$$

$$L_2 : \frac{x+1}{2} = \frac{y-0}{1} = \frac{z-0}{1} = \mu$$

Hence any point on L_1 and L_2 can be

$$(\lambda, \lambda-1, \lambda) \text{ and } (2\mu-1, \mu, \mu)$$

$$\therefore \frac{2\mu-1-\lambda}{2} = \frac{\mu-\lambda+1}{1} = \frac{\mu-\lambda}{1}$$

solving $\mu = 1$ and $\lambda = 3$

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$$\text{Hence } \frac{V_s}{V_\ell} = \frac{1}{27} = \frac{m}{n} \text{ or } \frac{n}{27} = \frac{m}{1} = k$$

$\therefore m$ and n are relatively prime

$$\Rightarrow k = 1, (m+n) = 28$$

11. Ans (B)

$$(I) \cos^{-1} \sqrt{x} + \cos^{-1} \sqrt{1-x} + \cos^{-1} \sqrt{1-y} = \frac{2\pi}{3}$$

$$\because 0 \leq x \leq 1 \text{ \& } \cos^{-1} \sqrt{x} = \sin^{-1} \sqrt{1-x}$$

$$\Rightarrow \frac{\pi}{2} + \cos^{-1} \sqrt{1-y} = \frac{2\pi}{3} \Rightarrow y = \frac{1}{4}$$

$$\Rightarrow 0 \leq 4x \leq 4 \text{ \& } 4y = 1$$

$$\text{or } 1 \leq 4x + 4y \leq 5$$

$$(II) \min(x^2 + 4x + 7) = 3$$

$$\max \left\{ \frac{\pi^2}{4}, 3 \right\} = 3$$

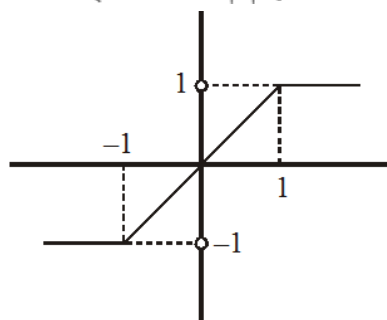
$$\lim_{z \rightarrow 0} \left[3 \cdot \frac{\sin^{-1} z}{z} \right] = 3$$

$$(III) |\sin x - \cos x| = \sin 2x - 1 \Rightarrow \sin 2x \geq 1$$

$$\Rightarrow \sin 2x = 1 \text{ only}$$

$$\Rightarrow \text{In } [0, 2\pi], x = \frac{\pi}{4} \text{ \& } \frac{5\pi}{4} \text{ are solutions.}$$

$$(IV) f(x) = \begin{cases} \text{sgn}(x) & |x| > 1 \\ x & |x| \leq 1 \end{cases}$$



12. Ans (D)

$$(I) \cos^{-1} \sqrt{x} + \cos^{-1} \sqrt{1-x} + \cos^{-1} \sqrt{1-y} = \frac{2\pi}{3}$$

$$\because 0 \leq x \leq 1 \text{ \& } \cos^{-1} \sqrt{x} = \sin^{-1} \sqrt{1-x}$$

$$\Rightarrow \frac{\pi}{2} + \cos^{-1} \sqrt{1-y} = \frac{2\pi}{3} \Rightarrow y = \frac{1}{4}$$

$$\Rightarrow 0 \leq 4x \leq 4 \text{ \& } 4y = 1$$

$$\text{or } 1 \leq 4x + 4y \leq 5$$

$$(II) \min(x^2 + 4x + 7) = 3$$

$$\max \left\{ \frac{\pi^2}{4}, 3 \right\} = 3$$

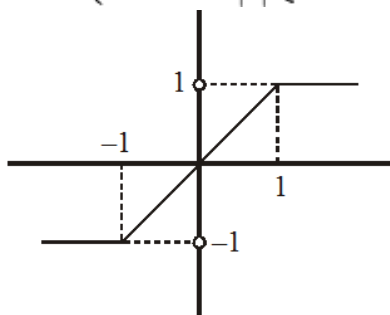
$$\lim_{z \rightarrow 0} \left[3 \cdot \frac{\sin^{-1} z}{z} \right] = 3$$

$$(III) |\sin x - \cos x| = \sin 2x - 1 \Rightarrow \sin 2x \geq 1$$

$$\Rightarrow \sin 2x = 1 \text{ only}$$

$$\Rightarrow \text{In } [0, 2\pi], x = \frac{\pi}{4} \text{ \& } \frac{5\pi}{4} \text{ are solutions.}$$

$$(IV) f(x) = \begin{cases} \operatorname{sgn}(x) & |x| > 1 \\ x & |x| \leq 1 \end{cases}$$



SECTION-II

1. Ans (3.00)

$f(x) = x^2 + ax^2 + bx + c = 0$ has 3 distinct integral roots and $t^3 + at^2 + bt + c = 0$ has no real roots where $t = (x+1)^2 + 1$ i.e. $t \geq 1$

Now $x^3 + ax^2 + bx + c = 0$ should have all integral roots less than equal to zero

$$x_1 \leq 0, x_2 \leq -1, x_3 \leq -2$$

$$\Rightarrow x_1 + x_2 + x_3 = -a \Rightarrow a \geq 3$$

2. Ans (0.00)

Eliminating y from two equations,

We get

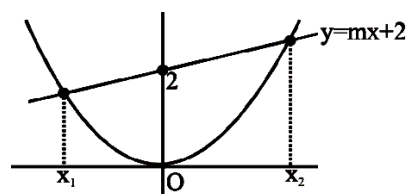
$$2mx + 4 = x^2$$

$$\Rightarrow x^2 - 2mx - 4 = 0$$

x_1 and x_2 are roots of this quadratic equation

$$\therefore x_1 + x_2 = 2m \text{ and } x_1 x_2 = -4$$

Now,

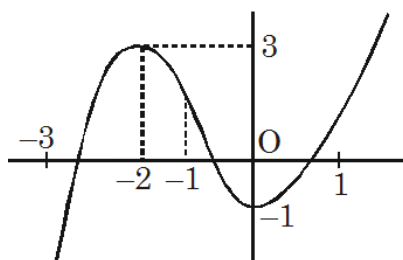


$$\begin{aligned} & \int_{x_1}^{x_2} \left(mx + 2 - \frac{x^2}{2} \right) dx \\ &= \frac{m}{2} (x_2^2 - x_1^2) + 2(x_2 - x_1) - \frac{1}{6} (x_2^3 - x_1^3) \\ &= (x_2 - x_1) \left[\frac{m}{2} (x_2 + x_1) + 2 - \frac{1}{6} (x_1^2 + x_1 x_2 + x_2^2) \right] \\ &= \sqrt{(x_2 + x_1)^2 - 4x_1 x_2} \left[\frac{m}{2} (x_2 + x_1) + 2 - \frac{1}{6} (x_2 + x_1^2 - x_2 x_1) \right] \\ &= \sqrt{4m^2 + 16} \left[\frac{m^2 + 4}{3} \right] \end{aligned}$$

Clearly above is minimum if $m = 0$

3. Ans (7.00)

Graph of $f(x)$ is



4. Ans (14.00)

$$a + c - 2b = 0$$

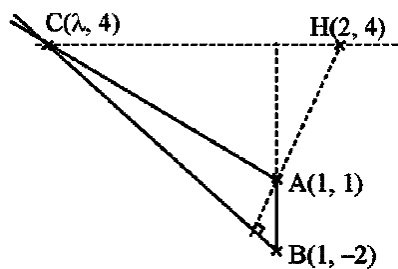
$\Rightarrow ax + by + c = 0$ passes through fixed point

$$(1, -2) \Rightarrow B \equiv (1, -2)$$

Now we plot the triangle on coordinate axes.

Clearly, H and C will have same y coordinate

$\therefore$ Let $C(\lambda, 4)$



$$\text{Slope of BC} = -\frac{2-1}{4-1} = \frac{4+2}{\lambda-1}$$

$$\lambda - 1 - 18 \Rightarrow \lambda = -17$$

$$C \equiv (-17, 4) \equiv (h, k)$$

$$\therefore 2h + 12k = -34 + 48 = 14$$

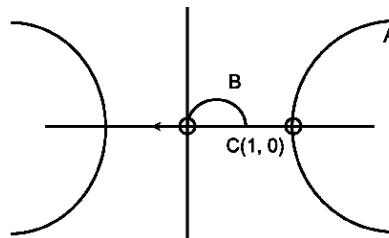
5. Ans (0.50)

$$f'_+(1) = \lim_{h \rightarrow 0} \frac{f(1+h) - f(1)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\frac{\pi}{4} + \frac{1+h-1}{2} - \frac{\pi}{4}}{h} = \frac{1}{2}$$

6. Ans (0.00)

A, B, C can be shown geometrically as below.



Clearly $A \cap B \cap C = \phi$